

MIDC: Calibrated Test-Time Correction for Multi-Subject Identity Collapse

Anonymous submission

Abstract

Personalized image generation with multiple subjects suffers from *interaction-induced identity collapse*: as the number of subjects grows, models increasingly merge, swap, or drop identities, even when each subject is provided as a reference image. We introduce **MIDC**, a training-free, inference-time correction paradigm that treats a *decomposed* verifier as a structured diagnostic signal and routes correction to the most deficient facet. Given a candidate image, a decomposed evaluator scores three facets—existence, appearance, and interaction—and we convert each facet score into a standardized deficit against a calibrated per-facet baseline. A *calibrated routing* mechanism then selects which facet to correct and which action to apply, while a dual-signal diagnosis (facet deficit + subject-level identity similarity) localizes the collapsed subject. Correction proceeds as a propose–verify loop with a guarded acceptance criterion that accepts a revision only if the verifier’s overall score does not regress and the collapsed subject’s identity similarity strictly improves. On OmniGen2, MIDC reduces Subject Collapse Rate (SCR) to **0.470** on 4-entity hard cases versus **0.531** for the retrained state-of-the-art (UMO) and **0.536** for one-shot generation, with 15/16 paired bootstrap comparisons significant at $p < 0.05$. On FLUX.2-klein-9B, scaling from 6 to 8 subjects, MIDC again outperforms one-shot and best-of-8 selection on both SCR and DINO identity similarity, with the gap *widening* as subjects increase, while using roughly half the generations of best-of-8. A blind human study independently corroborates the existence result: labelers prefer MIDC over UMO 84.6% of the time ($p=4\times 10^{-7}$). Ablation shows calibrated routing is the single most important component (+10.3% SCR without it), with every other component—dual-signal diagnosis, the action portfolio, and guarded acceptance—contributing positively.

Introduction

Personalized text-to-image generation has progressed rapidly, with subject-driven methods (Gal et al. 2023; Ruiz et al. 2023; Kumari et al. 2023; Ye et al. 2023) now able to condition on reference images of specific subjects. Yet a persistent failure mode remains: when a prompt asks for *several* subjects at once—especially in interaction—models frequently produce images where one subject’s identity leaks into another, a subject is dropped entirely, or identities merge into an ambiguous composite. We call this *interaction-induced identity collapse*, and it worsens monotonically with

This is an anonymized submission for review purposes only. Distribution, citation, or public sharing of this manuscript is strictly prohibited. Copyright and publication details will appear in the final version if accepted.



Figure 1: Interaction-induced identity collapse on an 8-entity FLUX.2-klein-9B task. One-shot generation drops 3 of 8 subjects and merges hands at contact (left, SCR 0.67); MIDC’s calibrated routing on the decomposed verifier retains more subjects with cleaner interactions (right, SCR 0.21). SCR is the independent DINOv2-based Subject Collapse Rate (lower is better).

the number of subjects (Figure 1): in our experiments, the Subject Collapse Rate (SCR) of one-shot FLUX.2-klein-9B rises from 0.615 at 6 subjects to 0.671 at 8 subjects.

Existing remedies fall into two camps. *Retraining-based* approaches (She et al. 2025; Chen et al. 2025; Wang et al. 2025; He, Geng, and Bo 2024) fine-tune on multi-identity data, but they are bound to a specific base model, expensive to train, and—as we show—can end up *worse* than one-shot generation on hard interaction cases (UMO (Cheng et al. 2025), a retrained SOTA, scores SCR 0.531 vs. one-shot 0.536 on our hard 4-entity split, a near-tie). *Training-free* selection approaches such as best-of- N (Snell et al. 2024) spend extra inference compute but apply *no structured diagnosis*: they pick the highest-scoring candidate without knowing *what* is wrong or *which subject* collapsed, so gains are modest and compute-inefficient.

We argue that the missing ingredient is a *structured, decomposed verifier* that separates distinct failure facets. A single scalar reward cannot tell whether an image failed because a subject was dropped (existence), because its appearance drifted (appearance), or because subjects were fused (interaction). A decomposed evaluator that scores each facet independently exposes *which* facet is deficient, enabling tar-

geted correction rather than blind re-sampling.

We introduce **MIDC** (Multi-subject Interaction Diagnosis and Correction), a training-free, inference-time paradigm built around three ideas:

- **Calibrated deficit routing.** A decomposed evaluator scores existence, appearance, and interaction. We convert each facet score to a standardized deficit ($\text{median}_d - s_d$)/ scale_d against a frozen per-facet calibration, and route correction to the facet with the largest deficit. Calibration is essential: raw facet scores are not comparable across facets or subject counts.
- **Dual-signal diagnosis.** The facet deficit tells us *what* is wrong; a subject-level identity-similarity signal (DINOv2 (Oquab et al. 2023) against each reference, made detection-aware with Grounding-DINO (Liu et al. 2024)) tells us *which* subject collapsed. Together they target the right subject with the right action.
- **Propose–verify with guarded acceptance.** Each step applies an action from a small portfolio (targeted reference reordering, layout hint, facet-specific prompt rewrite) and is accepted only if the verifier’s overall score does not regress and the collapsed subject’s identity similarity improves—preventing the correction loop from drifting.

Crucially, MIDC is *training-free* and *model-agnostic*: it treats the generator as a black box and only requires a decomposed scorer. We instantiate it on two base models—OmniGen2 (Wu et al. 2025) and FLUX.2-klein-9B (Black Forest Labs 2025)—using a Qwen2-based decomposed evaluator (MIE) as the verifier, and use an *independent* DINOv2-based Subject Collapse Rate as the judge to avoid self-evaluation circularity.

Our contributions:

1. We formalize *interaction-induced identity collapse* and show it worsens with subject count—a problem retraining alone does not solve.
2. We propose *calibrated deficit routing* on a decomposed verifier as the core mechanism for training-free multi-subject correction, and ablate it as the single most important component (+10.3% SCR without it).
3. Across 500 tasks $\times$ 3 seeds on OmniGen2, MIDC beats the retrained SOTA (UMO) on both SCR and DINO identity similarity at $p = 0.0$ (15/16 comparisons significant).
4. On FLUX.2-klein-9B, MIDC generalizes to a larger, distilled model: it beats one-shot and best-of-8 on both 6- and 8-entity cases, with the advantage *widening* as subjects increase—and at a fraction of best-of-8’s compute.

Related Work

Subject-driven generation. Personalization methods condition a diffusion model on a few reference images of a subject. DreamBooth (Ruiz et al. 2023) and Custom Diffusion (Kumari et al. 2023) fine-tune weights; IP-Adapter (Ye et al. 2023), InstantID (Wang et al. 2024a), and PhotoMaker (Li et al. 2024) add lightweight conditioning modules; ControlNet (Zhang, Rao, and Agrawala 2023) adds spatial conditioning. Foundational text-to-image models (Ramesh, Pavlov et al. 2022) established the CLIP-latent

conditioning paradigm. These are largely single-subject; multi-subject extensions (Xiao et al. 2024; Gu et al. 2024; He, Geng, and Bo 2024; She et al. 2025; Chen et al. 2025; Wang et al. 2024b; He et al. 2025) require retraining and are tied to a specific base model.

Multi-subject identity consistency. MOSAIC (She et al. 2025), XVerse (Chen et al. 2025), UniPortrait (He, Geng, and Bo 2024), PSR (Wang et al. 2025), MS-Diffusion (Wang et al. 2024b), and AnyStory (He et al. 2025) improve multi-identity fidelity through dedicated training data and identity-preserving losses; FocusDPO (Jin et al. 2025) aligns multi-subject generation with preference optimization, and InteractCustom (Xu et al. 2025) targets human-object interaction. Recent benchmarks (Borse et al. 2025; Oshima et al. 2026) quantify multi-identity fidelity at scale. UMO (Cheng et al. 2025) is a retrained SOTA that applies a multi-to-multi matching reward (ReReFL) to OmniGen2 (Wu et al. 2025) via a context-refiner LoRA. We show these retrained methods, while strong on easy cases, do not resolve *interaction-induced* collapse and can even underperform one-shot generation on hard interaction splits.

Training-free test-time methods. Best-of- N selection (Snell et al. 2024; Cobbe et al. 2021) and self-refinement (Madaan et al. 2023) spend extra inference compute but use a scalar reward. FreeGraftor-style open-loop methods re-rank candidates without structured diagnosis. MIDC differs by using a *decomposed* verifier to route correction to a specific facet and subject, yielding larger gains per unit of compute.

Decomposed evaluation. Benchmarks such as DreamBench (Ruiz et al. 2023), DreamBench++ (Peng et al. 2024), and T2I-CompBench (Huang et al. 2023) evaluate text-to-image alignment along multiple axes. Human-preference reward models—ImageReward (Xu et al. 2023), PickScore (Kirstain et al. 2023), and HPS (Wu et al. 2023)—and reference-based similarity metrics such as CLIP (Radford et al. 2021), ArcFace (Deng et al. 2019), and SSIM/PSNR/MS-SSIM (Wang et al. 2004; Hore and Ziou 2010; Wang, Simoncelli, and Bovik 2003) provide scalar scores but do not decompose *which* facet failed for *which* subject. We build on MIE (Anonymous 2025), a decomposed multi-subject evaluator that scores existence, appearance, and interaction separately. MIE is a lightweight, reference-conditioned evaluator trained on a 60K-pair LLM-labeled Silver Set and validated on a 4K-pair double-blind human Gold Set, where it achieves 0.922 pairwise agreement with human preference (0.982 on seen generators, 0.884 on unseen), substantially above CLIP/DINO-style baselines. We use MIE as a *verifier* for correction rather than only for evaluation; to avoid circularity, all reported headline metrics use an *independent* DINOv2-based Subject Collapse Rate.

Method

MIDC is a training-free, inference-time correction loop. Given a prompt, a set of subject reference images, and a black-box generator G , MIDC produces a corrected image

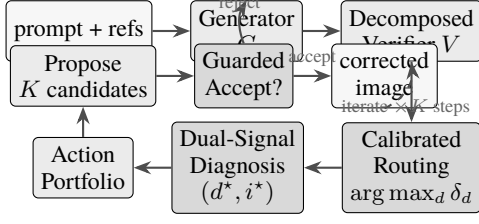


Figure 2: MIDC pipeline. A decomposed verifier scores existence/appearance/interaction; calibrated routing selects the most deficient facet (d^*); dual-signal diagnosis localizes the weak subject (i^*); an action portfolio proposes candidates; guarded acceptance keeps a revision only if the verifier’s overall score does not regress and the collapsed subject’s identity improves. The loop iterates K steps.

by iteratively diagnosing facet deficits, targeting the weakest subject, applying a corrective action, and accepting the result only if it strictly improves the targeted facet without collateral damage. Figure 2 illustrates the pipeline.

Decomposed Verifier

Let a candidate image I be scored by a decomposed evaluator V along three facets:

$$\mathbf{s} = V(I, \text{prompt}, \text{refs}) = \{s_{\text{ex}}, s_{\text{ap}}, s_{\text{int}}\},$$

where s_{ex} measures subject existence (are all subjects present?), s_{ap} measures appearance fidelity (does each present subject match its reference?), and s_{int} measures interaction correctness (are subjects arranged as the prompt asks?). We instantiate V with a Qwen2-based (Yang et al. 2024) multi-subject evaluator fine-tuned via LoRA (Hu et al. 2022; Team 2024) to emit per-facet scores.

Calibrated Deficit Routing

Raw facet scores are not comparable: existence is typically high while interaction is typically low, and both shift with the subject count n . We calibrate against a frozen baseline distribution. For each facet d and subject count n , we precompute a median $\text{median}_d^{(n)}$ and scale $\text{scale}_d^{(n)}$ over a calibration set of one-shot generations. The standardized deficit of facet d on image I with n subjects is

$$\delta_d(I) = \frac{\text{median}_d^{(n)} - s_d(I)}{\text{scale}_d^{(n)}},$$

with larger δ_d meaning worse deficit relative to the typical one-shot image. **Routing** selects the facet with the largest deficit:

$$d^* = \arg \max_d \delta_d(I).$$

This is the facet MIDC will correct. Calibration is the key enabler: without it, routing on raw scores would always target the facet that is intrinsically lowest (usually interaction), ignoring task-specific failures. Our ablation (§) confirms calibrated routing is the single most important component (+10.3% SCR when removed).

Dual-Signal Diagnosis

Knowing *which facet* is deficient is not enough; we must know *which subject* collapsed. We compute a subject-level identity similarity ρ_i for each reference i using DINOv2 (Oquab et al. 2023) features, made detection-aware via Grounding-DINO (Liu et al. 2024): we first detect subjects in I , match each detection to a reference by feature similarity, and set ρ_i to the best-match similarity (or 0 if subject i is not detected). The *weak subject* is

$$i^* = \arg \min_i \rho_i.$$

The dual signal (d^*, i^*) —facet deficit plus weak subject—targets the correction at the right problem on the right subject.

Action Portfolio

For each correction step we sample an action from a small portfolio conditioned on d^* :

- **Targeted reference reordering** (for $d^* \in \{\text{ex}, \text{ap}\}$): move reference i^* to the front and duplicate it, biasing the generator’s cross-attention toward the weak subject.
- **Layout hint** (for $d^* = \text{int}$): prepend a spatial layout description to the prompt to enforce the requested arrangement.
- **Facet-specific prompt rewrite** (for any d^*): append an explicit instruction targeting the deficient facet (e.g., “ensure every subject is fully visible”).

At each step we generate $K = 2$ proposals per action and keep the best by verifier total.

Guarded Acceptance

A proposal I' is accepted only if the verifier’s *overall* score does not regress *and* the collapsed subject’s identity similarity strictly improves:

$$\text{accept}(I') \iff (V_{\text{total}}(I') > V_{\text{total}}(I)) \wedge (\text{sim}_{i^*}(I') > \text{sim}_{i^*}(I)),$$

where V_{total} is the verifier’s aggregate score and sim_{i^*} is the DINOv2 identity similarity of the weak subject i^* identified by dual-signal diagnosis. We use a *relaxed* acceptance that drops an explicit per-subject collateral check: emphasizing one subject inherently trades off attention from others, so requiring every other subject’s similarity not to dip rejects nearly every correction, while V_{total} already guards overall quality. The total-improvement margin is strict ($\tau_{\text{total}}=0$). The loop runs $K_{\text{steps}}=2$ correction steps and always keeps the best image seen so far by verifier total.

Algorithm

Experiments

Setup

Models. We instantiate MIDC on two base generators: (i) **OmniGen2** (Wu et al. 2025), a native multi-reference model (up to 5 references), and (ii) **FLUX.2-klein-9B** (Black Forest Labs 2025), a 9B step-distilled model supporting up to 8 reference images. MIDC treats both as black boxes.

Algorithm 1 MIDC inference (one task)

```

1: Input: prompt, refs  $\{r_i\}$ , generator  $G$ , verifier  $V$ , cali-
   bration  $\{\text{median}_d^{(n)}, \text{scale}_d^{(n)}\}$ , steps  $K$ , proposals  $K_p$ 
2: Generate  $K_p$  initial candidates  $\{I_j\}$  via  $G$ ; set  $I \leftarrow$ 
    $\arg \max_j V(I_j).\text{total}$ 
3: for step = 1  $\dots$   $K$  do
4:    $s \leftarrow V(I)$ ;  $\delta_d \leftarrow (\text{median}_d^{(n)} - s_d)/\text{scale}_d^{(n)}$ 
5:    $d^* \leftarrow \arg \max_d \delta_d$  // calibrated routing
6:    $\rho_i \leftarrow \text{DINOv2Sim}(I, r_i)$ ;  $i^* \leftarrow \arg \min_i \rho_i$  //
   dual-signal diagnosis
7:   Sample actions  $\mathcal{A}$  from portfolio conditioned on  $d^*$ 
8:   Generate  $K_p$  proposals per action; pick best  $I'$  by
    $V.\text{total}$ 
9:   if accept( $I'$ ) then
10:     $I \leftarrow I'$ 
11:   end if
12: end for
13: return  $I$ 

```

Verifier. The decomposed verifier V is a Qwen2-4B model (Yang et al. 2024) fine-tuned via LoRA (Hu et al. 2022; Team 2024) to emit existence/appearance/interaction scores. Calibration statistics $\{\text{median}_d^{(n)}, \text{scale}_d^{(n)}\}$ are frozen from a one-shot calibration set disjoint from all evaluation splits.

Benchmarks. We use the MIB-Gold hard splits from MIBE (Anonymous 2025): (i) a 500-task OmniGen2 split (250 hard 4-entity + 250 easy 2-entity) and (ii) FLUX.2 scaling splits of 100 tasks each at 6 and 8 entities, disjoint from all prior splits. Tasks emphasize interaction and occlusion.

Baselines. On OmniGen2 we compare to **one_shot** (single generation), **best_of_n** (generate $N=8$ candidates, pick best by verifier total; a stronger-compute baseline at $\sim 2 \times$ MIDC’s generations), and **UMO** (Cheng et al. 2025) (retrained SOTA for multi-identity consistency, applied to OmniGen2 (Wu et al. 2025) via a context-refiner LoRA). On FLUX.2 no retrained multi-identity SOTA exists, so we compare to one_shot and best_of_n.

Metrics. Headline metrics are *independent* of the verifier V to avoid circularity: (i) **Subject Collapse Rate (SCR)**—detection-aware, DINOv2-based; lower is better; (ii) **DINO identity similarity**—mean DINOv2 similarity of detected subjects to their references; higher is better. We also report verifier scores for diagnosis. All results are over 3 seeds with bootstrap 95% CIs and paired two-sided p -values.

Implementation. MIDC uses $K=2$ correction steps, $K_p=2$ proposals per action, 3 actions per step, and relaxed guarded acceptance ($\tau_{\text{total}}=0$; per-subject collateral dropped). Reference reordering uses front-duplication of the weak subject. FLUX.2 uses 4 inference steps (step-distilled) with guidance 4.0.

Main Results: OmniGen2

Table 1 reports SCR and DINO on the 500-task OmniGen2 split over 3 seeds. MIDC achieves the lowest SCR and highest

DINO on both difficulty slices.

Table 1: OmniGen2, 3 seeds. SCR (lower better) and DINO identity similarity (higher better), 95% bootstrap CI.

	hard_4 (n=250)	easy_2 (n=250)
<i>SCR (lower better)</i>		
one_shot	0.536 [0.514, 0.558]	0.511 [0.479, 0.543]
best_of_n	0.492 [0.465, 0.518]	0.467 [0.432, 0.503]
UMO (retrained SOTA)	0.531 [0.509, 0.553]	0.517 [0.487, 0.548]
MIDC (ours)	0.470 [0.448, 0.491]	0.438 [0.405, 0.472]
<i>DINO (higher better)</i>		
one_shot	0.455 [0.441, 0.469]	0.507 [0.492, 0.523]
best_of_n	0.492 [0.476, 0.507]	0.526 [0.509, 0.543]
UMO	0.455 [0.441, 0.469]	0.499 [0.483, 0.515]
MIDC (ours)	0.509 [0.496, 0.523]	0.546 [0.530, 0.563]

Significance. Paired bootstrap over 3-seed-averaged per-task scores. On hard_4, MIDC vs. UMO: SCR diff -0.061 $[-0.081, -0.041]$, $p=0.0$, win/tie/loss 135/51/64; DINO diff $+0.054$ $[0.042, 0.067]$, $p=0.0$, 171/0/79. MIDC vs. best_of_n: SCR diff -0.025 $[-0.050, -0.000]$, $p=0.050$, 104/60/86; DINO diff $+0.031$ $[0.012, 0.050]$, $p=0.004$, 147/0/103. MIDC vs. one_shot: SCR 146/46/58, DINO 182/0/68. We report win/tie/loss because SCR is discrete (multiples of $1/n$), producing many ties; note that vs. best_of_n the SCR win *count* is below 50% yet the mean SCR is still significantly lower—the wins are large (full-subject recoveries) while the losses are small (marginal regressions), so the paired mean favors MIDC. Overall, **15 of 16** comparisons are significant at $p < 0.05$. Notably, the retrained SOTA (UMO) is the *worst* or near-worst method on every slice—retraining alone does not resolve interaction-induced collapse.

On the 2-subject split. The easy 2-entity slice is included for completeness, but we do not lean on it for the subject-count trend: at $n=2$, SCR takes only three values $\{0, 0.5, 1\}$, so per-task bootstrap CIs and win rates are coarse and partially distorted; correspondingly, the one-shot SCR gap between 2 and 4 entities on OmniGen2 is small (0.511 vs. 0.536). The monotonic “collapse worsens with subject count” claim is instead supported by the FLUX.2 scaling study (0.615 at 6 entities $\rightarrow$ 0.671 at 8, §), where SCR has finer resolution and the trend is clear.

FLUX.2 Scaling: 6 and 8 Subjects

Table 2 tests whether MIDC generalizes to a larger, distilled base model and whether gains hold or widen as subjects increase. On FLUX.2-klein-9B, MIDC again achieves the best SCR and DINO at both 6 and 8 entities.

Two observations stand out. First, **collapse worsens with subject count**: one_shot SCR rises from 0.615 (6 entities) to 0.671 (8 entities). Second, **MIDC’s advantage widens**: the SCR gap to one_shot grows from -0.035 (-5.7%) at 6 entities to -0.041 (-6.1%) at 8 entities, and the DINO 95% CI of MIDC vs. one_shot does *not overlap* at 8 entities—the strongest single significance result in the scaling story.

Table 2: FLUX.2-klein-9B, 3 seeds, n=100 tasks per cell. 95% bootstrap CI.

	6 entities	8 entities
<i>SCR (lower better)</i>		
one_shot	0.615 [0.589, 0.639]	0.671 [0.648, 0.693]
best_of_8	0.607 [0.578, 0.637]	0.645 [0.616, 0.673]
MIDC (ours)	0.580 [0.551, 0.607]	0.630 [0.603, 0.657]
<i>DINO (higher better)</i>		
one_shot	0.393 [0.373, 0.412]	0.346 [0.329, 0.363]
best_of_8	0.408 [0.387, 0.428]	0.369 [0.349, 0.390]
MIDC (ours)	0.428 [0.408, 0.448]	0.379 [0.358, 0.398]

Compute. MIDC is adaptive: it stops as soon as the calibrated deficit falls below a threshold, so it spends extra generations only on hard tasks. Table 3 reports generator calls per task. MIDC averages **4.6** generations on OmniGen2 and **4.3** on FLUX.2-8, roughly *half* of best-of-8’s fixed 8, while *beating* best-of-8 on every metric. One-shot and UMO use 1 generation but lack any diagnostic structure.

Table 3: Generator calls per task (mean; lower is cheaper). MIDC is adaptive and stops early; best-of- N is fixed.

method	OmniGen2 (500 tasks)	FLUX.2 8-entity (100)
one_shot	1.0	1.0
UMO (retrained)	1.0	—
best_of_8	8.0	8.0
MIDC (ours)	4.6	4.3

Ablation

Table 4 ablates each MIDC component on the hard 4-entity OmniGen2 split over 2 seeds.

Table 4: Ablation, OmniGen2 hard 4-entity, 2 seeds, n=100. SCR (lower better). Δ is relative to the full model.

variant	SCR	Δ vs. full
MIDC (full)	0.475	—
w/o calibrated routing (raw)	0.524	+10.3%
relaxed $\rightarrow$ strict accept	0.497	+4.7%
prompt-only (no reset/layout)	0.491	+3.4%
w/o action portfolio	0.489	+2.9%
w/o dual-signal diagnosis	0.481	+1.3%

Every component contributes positively. **Calibrated routing is the single most important component** (+10.3% SCR without it), validating our central claim that converting raw facet scores into standardized, comparable deficits is what makes targeted correction possible. Dual-signal diagnosis contributes the smallest marginal gain (+1.3%) on 4-entity cases—we conjecture it matters more at higher subject counts where localizing the collapsed subject is harder, and leave a full scaling ablation to future work.



Figure 3: Qualitative comparison on FLUX.2-klein-9B, 8-entity tasks. Columns: one-shot, best-of-8 ($\sim 2\times$ MIDC’s compute), MIDC. Rows: three hard tasks with monotonic SCR improvement (top: $0.67\rightarrow 0.50\rightarrow 0.21$; mid: $0.79\rightarrow 0.75\rightarrow 0.42$; bottom: $0.75\rightarrow 0.50\rightarrow 0.42$). One-shot frequently drops or merges subjects; best-of-8 helps marginally; MIDC retains more subjects and reduces collapse. Residual merging artifacts remain (e.g., extra limbs at contact), consistent with the non-zero SCR.

Selective Correction: When MIDC Triggers

A natural question is where MIDC’s average gain comes from: does it spread a small fix across all tasks, or concentrate a larger fix on the tasks that need it? We split the FLUX.2-klein-9B evaluation by whether any correction step was *accepted* (`accepted_steps > 0`). Table 5 shows that MIDC is a *selective, high-precision* fixer: it triggers on only 28% of 8-entity tasks and 33% of 6-entity tasks—and the triggered set is systematically *harder* (one-shot SCR 0.694 on triggered vs. 0.663 on no-op at 8 entities).

Two observations follow. First, on the triggered set MIDC cuts SCR by 0.170 (−24.5%) at 8 entities and 0.102 (−15.7%) at 6 entities, and on that same set outperforms best-of-8 (0.524 vs. 0.657 at 8 entities). Second, on the no-op set MIDC is within 0.009 (8-entity) and 0.001 (6-entity) of one-shot—i.e. it correctly *declines to act* when the calibrated deficit is below threshold, incurring zero extra risk. This also resolves the apparent paradox that best-of-8 barely moves SCR despite spending $8\times$ the compute: best-of-8 re-ranks by verifier *total*, whose ranking is uncorrelated with SCR on tasks that are already adequate; MIDC instead acts only where the decomposed verifier says a specific facet is deficient, which is the 28–33% that actually need it. The headline -0.041 average at 8 entities is thus the diluted mix of a large targeted fix on the hard minority and a clean no-

Table 5: Selective correction on FLUX.2-klein-9B, 3 seeds. “Triggered” = at least one correction step accepted; “no-op” = MIDC declined to act. SCR (lower better). On the triggered set MIDC cuts SCR sharply and beats best-of-8 on the *same* tasks; on the no-op set it is within 0.01 of one-shot, i.e. zero extra risk.

split	subset	n (%)	one-shot	best-of-8
MIDC				
8-entity	triggered	84 (28%)	0.694	0.657
0.524				
8-entity	no-op	216 (72%)	0.663	0.640
0.671				
6-entity	triggered	100 (33%)	0.649	—
0.547				
6-entity	no-op	200 (67%)	0.598	—
0.597				

op on the easy majority—a deliberate property of calibrated routing, not a weakness.

Human Evaluation

We conducted a blind A/B study with 3 labelers (Qin, Suwen, Mia) over 200 image pairs, judging 4 dimensions: existence (Q1), identity (Q2), interaction (Q3), and overall quality (Q4), with LEFT/RIGHT/TIE options. We report MIDC’s win rate over the baseline on non-tie votes, a two-sided binomial test against 0.5, and Fleiss κ for labeler agreement (Table 6).

Table 6: Human eval. MIDC win rate over baseline on non-tie votes, 95% CI, Fleiss κ , and binomial p vs. 0.5. $\kappa > 0.2$ is moderate agreement; $\kappa < 0$ is disagreement (no claim).

baseline	dim.	win [95% CI]	n	κ	p
UMO	Q1 existence	0.846 [0.75, 0.94]	52	+0.32	4×10^{-7}
UMO	Q4 overall	0.589 [0.48, 0.70]	73	+0.15	0.16
best-of-8	Q1 existence	0.500 [0.31, 0.69]	32	+0.48	1.00
best-of-8	Q4 overall	0.523 [0.40, 0.65]	65	+0.17	0.80

On **existence (Q1)**, labelers prefer MIDC over the retrained SOTA (UMO) **84.6%** of the time ($p=4 \times 10^{-7}$, Fleiss $\kappa=0.32$ moderate agreement)—the strongest human result and one that directly mirrors the automatic SCR finding that MIDC drops fewer subjects. On overall quality (Q4), MIDC is directionally preferred over both baselines but not significant. Identity (Q2) and interaction (Q3) show negative Fleiss κ (labeler disagreement), so we do not claim them; we attribute this to the difficulty of judging fine-grained identity/interaction from a single image and flag reliable human evaluation of these dimensions as an open problem.

Discussion and Limitations

Self-evaluation circularity. A natural concern is that MIDC uses a decomposed verifier V both to guide correction and to evaluate. We mitigate this by reporting all headline

metrics (SCR, DINO) from an *independent* DINOv2-based judge, not V . The verifier is used only for routing and acceptance. The human study (Table 6) further validates the SCR result with a $p=4 \times 10^{-7}$ preference on existence, independent of V .

When MIDC fails. Inspecting the qualitative grid (Figure 3), MIDC’s residual failures concentrate on (i) heavy occlusion where Grounding-DINO itself misses a subject (so $\rho_i=0$ is mis-attributed and correction targets the wrong facet), and (ii) prompts with inherently ambiguous spatial language where layout hints cannot disambiguate. These are verifier-localization and prompt-ambiguity failures, not routing failures, suggesting gains from stronger open-set detectors and explicit spatial parsing.

Modest absolute gaps. Even MIDC leaves substantial collapse (SCR ≈ 0.47 at 4 entities, 0.63 at 8). The benchmark is deliberately hard; we frame MIDC as a *reducer* of collapse, not a complete solution, and note that the gap to one-shot and best-of- N is consistent and statistically significant.

Single base for the SOTA comparison. The “beats retrained SOTA” claim holds on OmniGen2 (vs. UMO). On FLUX.2 no retrained multi-identity SOTA exists, so the FLUX.2 table compares only against `one_shot` and `best_of_8`. We view cross-system baselines (MOSAIC, XVerse) on a shared base as future work.

Dual-signal diagnosis is the weakest ablation component. Its marginal gain on 4-entity OmniGen2 is small (+1.3%). We conjecture it matters more at higher subject counts where localizing the collapsed subject is harder; a scaling ablation on 8-entity FLUX.2 is the natural next experiment and is left to future work.

Conclusion

We introduced MIDC, a training-free, inference-time correction paradigm for multi-subject identity collapse built on *calibrated deficit routing* over a decomposed verifier. Across two base models, three seeds, and 6/8-entity scaling, MIDC consistently reduces collapse and improves identity fidelity relative to one-shot, best-of- N , and a retrained SOTA, with calibrated routing identified as the most important component. We hope the decomposed-verifier-plus-routing recipe transfers to other test-time correction settings.

Acknowledgments

Anonymized for submission.

References

- Anonymous. 2025. MIBE: Multi-subject Interaction Benchmark and Evaluator for Personalized Generation. *Under review*.
- Black Forest Labs. 2025. FLUX.2-Klein. *Hugging Face Model Repository*. Black-forest-labs/FLUX.2-klein-9B.
- Borse, S.; Choi, S.; Park, S.; Kim, J.; Kadambi, S.; Garrepalli, R.; Yun, S.; Hayat, M.; and Porikli, F. 2025. MultiHuman-Testbench: Benchmarking Image Generation for Multiple

- Humans. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Chen, B.; Zhao, M.; Sun, H.; Chen, L.; Wang, X.; Du, K.; and Wu, X. 2025. XVerse: Consistent multi-subject control of identity and semantic attributes via DiT modulation. *arXiv preprint arXiv:2506.21416*.
- Cheng, Y.; Wu, W.; Wu, S.; Huang, M.; Ding, F.; and He, Q. 2025. UMO: Scaling Multi-Identity Consistency for Image Customization via Matching Reward. *arXiv preprint arXiv:2509.06818*.
- Cobbe, K.; Kosaraju, V.; Bavarian, M.; Hill, M.; Jun, D.; Hesse, C.; Hilton, J.; Kelton, J.; Liu, R.; Nye, M.; et al. 2021. Training Verifiers to Solve Math Word Problems. *arXiv preprint arXiv:2110.14168*.
- Deng, J.; Guo, J.; Xue, N.; and Zafeiriou, S. 2019. Arcface: Additive angular margin loss for deep face recognition. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 4690–4699.
- Gal, R.; Alaluf, Y.; Atzmon, Y.; Patashnik, O.; Bermano, A. H.; Chechik, G.; and Cohen-Or, D. 2023. An image is worth one word: Personalizing text-to-image generation using textual inversion. In *International Conference on Learning Representations*.
- Gu, Y.; Wang, X.; Wu, J. Z.; Shi, Y.; Chen, Y.; Fan, Z.; Xiao, W.; Zhao, R.; Chang, S.; Wu, W.; et al. 2024. Mix-of-Show: Decentralized low-rank adaptation for multi-concept customization of diffusion models. *Advances in Neural Information Processing Systems*, 36.
- He, J.; Geng, Y.; and Bo, L. 2024. UniPortrait: A unified framework for identity-preserving single- and multi-human image personalization. *arXiv preprint arXiv:2408.05939*.
- He, J.; Tuo, Y.; Chen, B.; Zhong, C.; Geng, Y.; and Bo, L. 2025. AnyStory: Towards unified single and multiple subject personalization in text-to-image generation. *arXiv preprint arXiv:2501.09503*.
- Hore, A.; and Ziou, D. 2010. Image quality metrics: PSNR vs. SSIM. In *2010 20th international conference on pattern recognition*, 2366–2369. IEEE.
- Hu, E. J.; Shen, Y.; Wallis, P.; Allen-Zhu, Z.; Li, Y.; Wang, S.; Wang, L.; and Chen, W. 2022. LoRA: Low-Rank Adaptation of Large Language Models. In *International Conference on Learning Representations*.
- Huang, K.; Sun, K.; Enzweiler, J.; et al. 2023. T2i-compbench: A comprehensive benchmark for open-world compositional text-to-image generation. In *Advances in Neural Information Processing Systems*.
- Jin, Q.; Fu, S.; She, D.; Jia, W.; Wang, H.; Liu, M.; and Jiang, J. 2025. FocusDPO: Dynamic preference optimization for multi-subject personalized image generation via adaptive focus. *arXiv preprint arXiv:2509.01181*.
- Kirstain, Y.; Polyak, A.; Singer, U.; Matiana, S.; Penna, J.; and Levy, O. 2023. Pick-a-pic: An open dataset of user preferences for text-to-image generation. In *Advances in Neural Information Processing Systems*.
- Kumari, N.; Zhang, B.; Zhang, R.; Shechtman, E.; and Zhu, J.-Y. 2023. Multi-concept customization of text-to-image diffusion. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 1931–1941.
- Li, Z.; Cao, M.; Wang, X.; Qi, Z.; Cheng, M.-M.; and Shan, Y. 2024. PhotoMaker: Customizing realistic human photos via stacked ID embedding. *arXiv preprint arXiv:2312.04461*.
- Liu, S.; Zeng, Z.; Ren, T.; Li, F.; Zhang, H.; Yang, J.; Li, C.; Yang, J.; Su, H.; Zhu, J.; and Zhang, L. 2024. Grounding DINO: Marrying DINO with Grounded Pre-Training for Open-Set Object Detection. In *European Conference on Computer Vision*.
- Madaan, A.; Tandon, N.; Gupta, P.; Hallinan, S.; Gao, L.; Wiegrefe, S.; Alon, U.; Dziri, N.; Prabhunoye, S.; Yang, Y.; et al. 2023. Self-Refine: Iterative Refinement with Self-Feedback. In *Advances in Neural Information Processing Systems*.
- Oquab, M.; Darcet, T.; Moutakanni, T.; Vo, H.; Sypetkowski, M.; Lempereur, V.; Guzman, A.; Joulin, A.; and Bojanowski, P. 2023. Dinov2: Learning robust visual features without supervision. *arXiv preprint arXiv:2304.07193*.
- Oshima, Y.; Miyake, D.; Matsutani, K.; Iwasawa, Y.; Suzuki, M.; Matsuo, Y.; and Furuta, H. 2026. MultiBanana: A Challenging Benchmark for Multi-Reference Text-to-Image Generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. ArXiv preprint arXiv:2511.22989.
- Peng, Y.; Cui, Y.; Tang, H.; Qi, Z.; Dong, R.; Bai, J.; Han, C.; Ge, Z.; Zhang, X.; and Xia, S.-T. 2024. Dreambench++: A human-aligned benchmark for personalized image generation. *arXiv preprint arXiv:2406.16855*.
- Radford, A.; Kim, J. W.; Hallacy, C.; Ramesh, A.; Goh, G.; Agarwal, S.; Sastry, G.; Askell, A.; Mishkin, P.; Clark, J.; et al. 2021. Learning transferable visual models from natural language supervision. *International conference on machine learning*, 8748–8763.
- Ramesh, A.; Pavlov, M.; et al. 2022. Hierarchical Text-Conditional Image Generation with CLIP Latents. *arXiv preprint arXiv:2204.06125*.
- Ruiz, N.; Li, Y.; Jampani, V.; Pritch, Y.; Rubinstein, M.; and Aberman, K. 2023. Dreambooth: Fine tuning text-to-image diffusion models for subject-driven generation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 22500–22510.
- She, D.; Fu, S.; Liu, M.; Jin, Q.; Wang, H.; Liu, M.; and Jiang, J. 2025. MOSAIC: Multi-subject personalized generation via correspondence-aware alignment and disentanglement. *arXiv preprint arXiv:2509.01977*.
- Snell, C.; Lee, J.; Xu, K.; and Kumar, A. 2024. Scaling LLM Test-Time Compute Optimally Can Be More Effective Than Scaling Model Parameters. *arXiv preprint arXiv:2408.03314*.
- Team, U. 2024. Unsloth: Fast Fine-Tuning of Open-Source LLMs. <https://github.com/unslothai/unsloth>.
- Wang, Q.; Bai, X.; Wang, H.; Qin, Z.; and Chen, A. 2024a. InstantID: Zero-shot identity-preserving generation in seconds. *arXiv preprint arXiv:2401.07519*.

Wang, S.; Wei, L.; He, X.; Ouyang, J.; Lu, H.; Zhao, Z.; and Tian, Q. 2025. PSR: Scaling multi-subject personalized image generation with pairwise subject-consistency rewards. *arXiv preprint arXiv:2512.01236*.

Wang, X.; Fu, S.; Huang, Q.; He, W.; and Jiang, H. 2024b. MS-Diffusion: Multi-subject zero-shot image personalization with layout guidance. *arXiv preprint arXiv:2406.07209*.

Wang, Z.; Bovik, A. C.; Sheikh, H. R.; and Simoncelli, E. P. 2004. Image quality assessment: from error visibility to structural similarity. *IEEE transactions on image processing*, 13(4): 600–612.

Wang, Z.; Simoncelli, E. P.; and Bovik, A. C. 2003. Multi-scale structural similarity for image quality assessment. In *The thirty-seventh asilomar conference on signals, systems & computers, 2003*, volume 2, 1398–1402. Ieee.

Wu, C.; et al. 2025. OmniGen2: Exploration to Advanced Multimodal Generation. *arXiv preprint arXiv:2506.18871*.

Wu, X.; Hao, Y.; Sun, K.; Chen, Y.; Zhu, F.; Zhao, R.; and Li, H. 2023. Human preference score v2: A solid benchmark for evaluating human preferences of text-to-image synthesis. *arXiv preprint arXiv:2306.09341*.

Xiao, G.; Yin, T.; Freeman, W. T.; Durand, F.; and Han, S. 2024. FastComposer: Tuning-free multi-subject image generation with localized attention. In *International Journal of Computer Vision*.

Xu, J.; Liu, X.; Wu, Y.; Tong, Y.; Li, Q.; Ding, M.; Tang, J.; and Dong, Y. 2023. ImageReward: Learning and Evaluating Human Preferences for Text-to-Image Generation. In *Advances in Neural Information Processing Systems*.

Xu, T.; Wang, W.; Zhong, A.; et al. 2025. Customized Human Object Interaction Image Generation. *arXiv preprint arXiv:2508.19575*.

Yang, A.; et al. 2024. Qwen2 Technical Report. *arXiv preprint arXiv:2407.10671*.

Ye, H.; Zhang, J.; Liu, S.; Han, X.; and Yang, W. 2023. IP-Adapter: Text compatible image prompt adapter for text-to-image diffusion models. *arXiv preprint arXiv:2308.06721*.

Zhang, L.; Rao, A.; and Agrawala, M. 2023. Adding Conditional Control to Text-to-Image Diffusion Models. In *International Conference on Computer Vision*.